Number operation properties

1 State whether each expression is an example of the associative law, commutative law, reordering or distributive law.

a $15 + 5 = 5 + 15$

b $24 \times 6 = 6 \times 24$

c $54 \div 9 = 9 \div 54$

d $24 + 6 + 2 = 24 + (6 + 2)$

e $126 \div 3 = (120 + 6) \div 3$

f $(15 + 6) \times 4 = (15 \times 4) + (6 \times 4)$

Associative law

2 State whether the following statements are true or false.

a $4 \times (3 + 5) = 4 \times 3 + 4 \times 5$

b $4 + (3 \times 5) = 4 \times 3 + 4 \times 5$

c $(4 \times 3) + 5 = 4 \times 5 + 3 \times 5$

d $(4 + 3) \times 5 = 4 \times 5 + 3 \times 5$

3 Fill in the boxes to complete the working out using the associative law.

a $35 + 29 + 11 = 35 + (\square + \square)$

b $28 + 46 - 16 = 28 + (\square - \square)$

4 Consider $27 + 48 + 13$.

a Which pair of numbers will be easiest to add together first?

b Fill in the boxes to complete the working out.
$27 + 48 + 13 = 27 + (\square + \square)$

5 Fill in the boxes to complete the working out using the associative law.
$13 \times 4 \times 5 = \square \times (4 \times \square)$

Reordering

6 Evaluate $17 + 39 + 23$ using reordering.

7 Consider $52 - 24 - 12$.

a Is $52 - 12 - 24$ equal to the above expression?

b Which of the arithmetic rules can we apply to the expression to transform it into $52 - 12 - 24$?

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8 Derek is evaluating the subtraction below. He noticed that it would be easier to subtract 27 from 47 first and rearranged the numbers using reordering.
Complete the following statement and then evaluate it:
$47 - 18 - 27 = \square - 27 - \square$

9 Consider $56 - 18 - 26$.

- **a** Which of the two numbers will be easier to subtract from 56 first?
- **b** Complete the following statement and then evaluate it:
$56 - 18 - 26 = 56 - \square - \square$

10 Evaluate $57 - 29 - 17$ using reordering.

11 Consider $48 \div 8 \div 2$.

- **a** Is $48 \div 2 \div 8$ equal to the above expression?
- **b** Which of the arithmetic rules can we apply to the expression to transform it into $48 \div 2 \div 8$?

Commutative Law

13 Consider $4 \times 13 \times 5$.

- **a** Which pair of numbers will be easiest to multiply together first?
- **b** Complete the following statement and then evaluate it:
$4 \times 13 \times 5 = \square \times \square \times 13$

14 Evaluate $4 \times 17 \times 5$ using regrouping.

Distributive Law

15 Consider 35×23.

- **a** Complete the following statement:
35×23 is the same as $35 \times (20 + \square)$.
- **b** Complete the following statement:
$35 \times (20 + 3)$ is the same as $35 \times \square + 35 \times 3$.
- **c** Which arithmetic rule explains the equality between 35×23 and $35 \times 20 + 35 \times 3$?
- **d** Evaluate the expression.

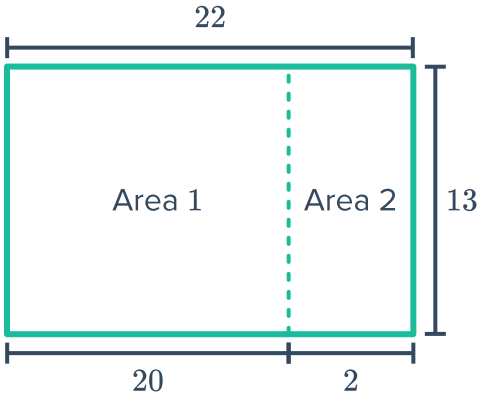
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16 Consider 26×29.

- **a** Complete the following statement:
  26×29 is the same as $26 \times (\square - 1)$.
- **b** Complete the following statement:
  $26 \times (30 - 1)$ is the same as $26 \times 30 - 26 \times \square$.
- **c** Which arithmetic rule explains the equality between 26×29 and $26 \times 30 - 26 \times 1$?
- **d** Evaluate the expression.

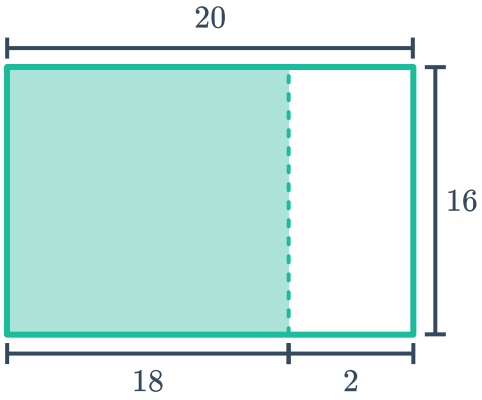
17 Consider the following rectangle:

- **a** Find the area of Area 1.
- **b** Find the area of Area 2.
- **c** Find the total area of the rectangle.



18 Consider the rectangle below:

- **a** Find the total area of the rectangle.
- **b** Find the non-shaded area of the rectangle.
- **c** Find the shaded area of the rectangle.



19 Evaluate the following using the distributive law.

- **a** 9×998
- **b** 7×1002
- **c** 11×124
- **d** 6×412

20 Dave is evaluating $132 \div 11 \div 4$. He noticed that it would be easier to divide 132 by 4 first and rearranged the numbers using reordering.
Fill in the boxes to complete his working out.
$132 \div 11 \div 4 = 132 \div \square \div 11$

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21 Consider $168 \div 7$.

a Fill in the box to complete the sentence.
$168 \div 7 = (140 + \square) \div 7$.

b Fill in the box to complete the sentence.
$(140 + 28) \div 7$ is the same as $140 \div 7 + \square \div 7$.

c Which arithmetic rule explains the equality between $168 \div 7$ and $140 \div 7 + 28 \div 7$?

d Evaluate the expression.

22 Consider $119 \div 7$.

a Fill in the box to complete the sentence.
$119 \div 7$ is the same as $(140 - \square) \div 7$.

b Fill in the box to complete the sentence.
$(140 - 21) \div 7$ is the same as $140 \div \square - 21 \div \square$.

c Which arithmetic rule explains the equality between $119 \div 7$ and $140 \div 7 - 21 \div 7$?

d Evaluate the expression.

23 Evaluate the following using the distributive law.

a $1616 \div 8$ **b** $396 \div 4$ **c** $522 \div 6$ **d** $585 \div 13$

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